Ansible Overwiew:

Ansible is an open-source automation tool used to manage IT infrastructure by automating tasks like configuration management, application deployment, and orchestration

What Ansible Does?

Automates repetitive tasks**: From installing packages to configuring services, Ansible handles it all.

Manages system configurations**: Ensures servers and environments stay consistent across deployments.

Deploys applications**: Supports rolling updates and zero-downtime deployments.

Orchestrates complex workflows**: Coordinates multi-tier applications and services across environments.

Key Features

Agentless architecture**: No need to install software on managed nodes; it uses SSH or WinRM.

Human-readable playbooks**: Written in YAML, playbooks describe desired system states in a clear, declarative format.

Idempotent execution**: Running the same playbook multiple times won’t change the system if it’s already in the desired state.

Extensible modules**: Thousands of built-in modules for tasks like file manipulation, service control, cloud provisioning, and more.

Why Engineers Love It?

Simple to learn**: No programming experience needed—just basic YAML and command-line skills.

Scales easily**: Works for small homelabs and massive enterprise environments alike.

Integrates with cloud platforms**: Supports AWS, Azure, GCP, and more for provisioning and automation.

Community-driven**: Backed by a vibrant open-source community and Red Hat’s enterprise support.

Real-World Use Cases

- Provisioning servers with specific OS and packages
- Enforcing security policies across fleets
- Deploying containerized apps with Kubernetes
- Automating CI/CD pipelines
- Managing network devices and firewalls

Thanks:

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